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RESEARCH ARTICLE

A vision transformer approach for fully automated and scalable dementia screening using clock drawing test images

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Abstract

INTRODUCTION: The clock drawing test (CDT) screens for dementia but requires trained scorers and lacks standardized criteria. Thus, we developed an automated vision transformer (ViT)-based diagnostic system with convolutional neural network preprocessing for analyzing hand-drawn CDT images.

METHODS: The architecture implements fine-tuned ViT feature extraction with linear classification for dementia prediction. Training used the National Health and Aging Trends Study (NHATS) dataset (n = 54,027), with testing on an independent clinical cohort from the Toronto Dementia Research Alliance (TDRA; n = 862; 522 dementia, 340 normal cognition).

RESULTS: The ViT approach predicted dementia with 76.5% balanced accuracy, outperforming human-scored features (74.3%) and existing deep learning models (MiniVGG = 73.3%, MobileNetV2 = 72.3%, relevance factor variational autoencoder = 69.1%) on the TDRA dataset.

DISCUSSION: This pen-and-paper compatible diagnostic system enables scalable remote cognitive screening through automated CDT image analysis that is competitive with human-scored features, potentially increasing diagnostic accessibility for diverse populations across varied socioeconomic contexts.

KEYWORDS
clock drawing test, cognitive assessment, deep learning, dementia screening, vision transformer

HIGHLIGHTS

- The vision transformer model achieves 76.5% accuracy in dementia detection from clock drawing tests, outperforming human scoring and existing deep learning methods.

Morris Freedman and Bradley R. Buchsbaum contributed equally to this work.

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- Novel convolutional neural network–based preprocessing automatically handles challenging image quality issues like shadows, irrelevant markings, and improper cropping.
- The system requires only a photo of a hand-drawn clock test, enabling scalable remote screening accessible across socioeconomic contexts.
- A feature-extraction model trained on 54,027 samples demonstrates robust generalization to an independent clinical dataset of 862 patients.
- This fully automated approach eliminates the need for trained scorers while maintaining diagnostic accuracy above manual methods.

1 | BACKGROUND

Dementia impacts millions worldwide, costing trillions annually, mostly in low/middle-income countries with limited diagnosis access.[1] This drives the need for affordable, scalable virtual screening tools.

The clock drawing test (CDT) is an effective dementia screening tool.[2,3] However, wider adoption is hindered by the need for trained clinicians to score drawings, the lack of standardized scoring methods, and difficulty in remote administration.

Recent work has addressed these limitations, using deep learning[4] for CDT scoring[5–7] or direct dementia diagnosis.[8–11] While diagnosis-based methods automatically discover relevant features for dementia diagnosis, they typically lack full automation (e.g., automated cropping) and direct comparative evaluation against feature extraction performed by humans and other automated methods on a common dataset. Furthermore, the architectures used for dementia diagnosis, including convolutional neural networks (CNN)s [9-10] and variational autoencoders (VAEs)s [11], either primarily focus on local features via limited receptive fields (CNN) or rely on specific latent space assumptions (VAE). We hypothesize that these characteristics may limit their ability to fully capture the holistic structure and complex spatial relationships within the CDT—features critical for reflecting cognitive status.

We propose a novel approach for automated dementia diagnosis from CDT images, using a pretrained vision transformer (ViT), "vit_l_16,"[12] to better capture critical global and relational features. Unlike prior models that predominantly used CNNs or VAEs for feature extraction, our method leverages ViT's self-attention mechanisms to process images globally, enabling the capture of long-range dependencies and spatial relationships between clock components. Our system integrates three key components: (1) a CNN-based preprocessor for automated clock identification and cropping from photographs, (2) a finetuned ViT trained on the National Health and Aging Trends Study dataset (NHATS; $n = 54{,}027$)[13] to generate predictive features, and (3) a linear model trained on an independent clinical dataset ($n = 862$) from the Toronto Dementia Research Alliance (TDRA) to predict dementia.

We compared our ViT-based approach against four benchmark methods on the TDRA clinical dataset: (1) human-scored features using the Freedman scoring system,[14] (2) features from modified MobileNetV2 (MNv2),[9,15,16] (3) features from MiniVGG,[10] and (4) features from a relevance factor variational autoencoder (RF-VAE;[11] see Table S1 in supporting information for a breakdown of the methods). These specific deep learning methods[9–11] were selected as comparators because they represent recent, relevant techniques designed for dementia diagnosis using only CDT image input without pre-drawn circles, providing a representative baseline. Our approach outperformed all four benchmark methods across all statistical measures evaluated.

Our method achieves greater-than-human performance using only standard pen-and-paper clock drawing images, providing an accessible, low-cost, standardized, and highly scalable screening tool suitable for remote deployment across varied socioeconomic contexts.

2 | METHODS

2.1 | Participants

We conducted a retrospective study using data from the TDRA and the NHATS. The study was approved by Clinical Trials Ontario (CTO).

2.1.1 | TDRA participants

The TDRA clinical research database, approved by CTO, consists of patients' demographic information, cognitive assessment data, and diagnoses provided by physicians from four memory clinics in Toronto located at Baycrest Health Sciences, Centre for Addiction and Mental Health (CAMH), Sunnybrook Health Sciences Centre, and University Health Network (UHN). Clinical history was sourced from either the patient, a care partner, medical records, or a combination. Data were obtained from August 2017 and August 2021, totaling 1986 samples with clock drawings.

2.1.2 | NHATS participants

Started in 2011, the NHATS, sponsored by the National Institute on Aging (grant number NIA U01AG032947) through a cooperative

RESEARCH IN CONTEXT

1. **Systematic Review**: A literature review using traditional sources (e.g., PubMed and preprint servers) revealed several approaches for automated clock drawing test (CDT) analysis. The vision transformer (ViT) architecture, while successful in general computer vision tasks, had not been previously applied to CDT analysis for dementia screening.
2. **Interpretation**: Our findings demonstrate that a ViT-based approach with convolutional neural network preprocessing can achieve superior dementia detection accuracy (76.5%) compared to human scoring and existing deep learning methods. The system's ability to analyze standard pen-and-paper clock drawings while outperforming human raters represents a significant advance in automated cognitive screening.
3. **Future Directions**: Key research priorities include: (a) comparing the system's diagnostic decisions against clinicians' qualitative clock drawing assessments, (b) investigating the integration of demographic data and additional cognitive tests to improve accuracy, (c) exploring model interpretability to understand which clock features drive predictions, and (d) validating performance across diverse populations and health-care settings.

agreement with the Johns Hopkins Bloomberg School of Public Health, is a longitudinal study of disability trends and trajectories among Medicare beneficiaries aged ≥ 65 years in the United States.[13] The anonymized publicly available NHATS database comprises annual interviews and cognitive assessments, with samples selected using a stratified random sampling of community-dwelling older adults. Data were obtained from rounds 1 to 11 (collected 2011–2021) from the NHATS website in February 2023, totaling 54,027 samples (57.1% female; age: mean 78.12, standard deviation 7.2). Samples from repeat assessments were used for model training. This approach leverages the large number of unique participants (> 5000) providing substantial inter-individual variability, while the prediction of time-varying outcomes (clock score and dementia status at the time of drawing) encourages models to learn state-dependent features rather than participant-specific ones.

2.2 | CDTs

2.2.1 | TDRA CDT

The CDT is included in both the Toronto Cognitive Assessment (TorCA) and the Behavioural Neurology Assessment-Short Form (BNA-SF).[14] During the assessment, participants were given an 8½ x 11–inch paper in portrait orientation and were instructed to "Draw the face of a clock, put in the numbers and set the hands at 10 after 11." The instructions were repeated to the patients as needed. As participants drew the clock, the administrator used an iPad to record the sequence of elements. The iPad provided options for different elements, allowing the administrator to select the element drawn by the patient as they drew the clock. After the patient completed the drawing, the administrator used the iPad's camera to capture an image of the clock drawing within the patient's TorCA or BNA-SF assessment, creating a digital copy.

The assessment was conducted by trained clinicians, including staff physicians in the memory clinic, clinical fellows, residents, medical students, certified psychological associates, nurses, and psychologists. In some cases, the TorCA or BNA-SF was administered by a research assistant.

The 15-point CDT scoring system[14] is shown in Table S2 in supporting information.

2.2.2 | NHATS CDT

CDT images were obtained from each participant by having them draw a picture of an analog clock with the hands showing 11:10 on a blank A4-size paper. CDT images were scored by the interviewers according to pre-defined criteria, using the range of 0 to 5 as follows: (0) not recognizable as a clock, (1) severely distorted depiction, (2) moderately distorted depiction, (3) mildly distorted depiction, (4) reasonably accurate depiction, and (5) accurate depiction.

2.3 | Clock image preprocessing

Clock images from the TDRA (≈ 960 × 1280, RGB) and NHATS (2560 × 3312, single channel) datasets, either scanned or photographed, exhibited significant quality variance. Preprocessing was essential to standardize these images for input into the "vit_l_16" dementia classification neural network, requiring a 512 × 512 resolution. Detailed preprocessing steps are provided in the Supplementary Methods in supporting information.

2.3.1 | Line filtering and preprocessing

For TDRA images, line filtering was applied to emphasize clock lines, converting images to black and white and removing irrelevant features like color and shadows. Pixel values were normalized (maximum 1, minimum 0), with values < 0.2 set to zero to reduce noise. NHATS images, pre-filtered, were inverted, and black rectangles censoring participant names were removed using a custom filter (see Supplementary Methods for details).

2.3.2 | Image cropping

To preserve clock drawing details, images were cropped before resizing to 512 × 512, as direct downsampling risked losing resolution. A

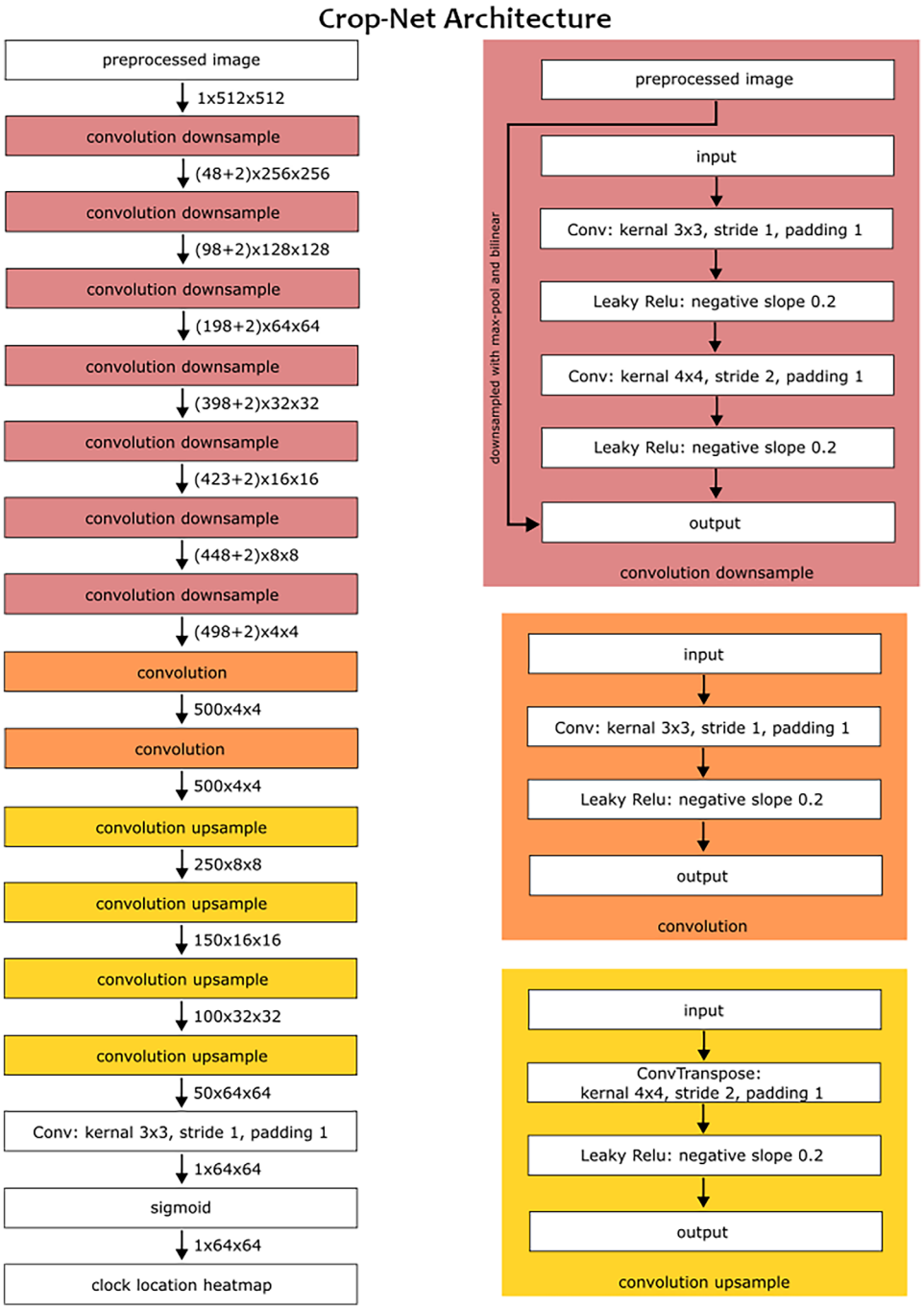

FIGURE 1 Crop-Net architecture. Network for locating clock drawings in an image. The full architecture is depicted on the left, with the component color-coded blocks expanded on the right. For the "Convolution downsample" block outputs, the numbers in brackets are the number of features with the first number referring to the last layer output and the "+2" referring to the downsampled (using max-pool and bilinear downsampling) image being added as two extra features. The number of features is increased relative to the input at the second convolution in the block, except for the first "Convolution downsample" block wherein the features are increased from 1 to 25 in the first convolution and 25 to 48 in the second

CNN, Crop-Net, was trained using PyTorch[17] to generate heatmaps identifying clock locations within images, addressing non-clock elements like written notes or paper edges. The network architecture is shown in Figure 1. Crop-Net was trained on the NHATS dataset with data augmentation (e.g., rotations, resizing, added noise, and lines; see Figure 2A), using mean squared error (MSE) as the loss function and the Adam algorithm[18] for optimization. A multi-step cropping script iteratively applied Crop-Net heatmaps to refine crops. Successfully cropped images from the NHATS and TDRA datasets were obtained after retraining Crop-Net on validated crops (see Figure 2B for TDRA examples). Full details, including training parameters and cropping iterations, are in the Supplementary Methods.

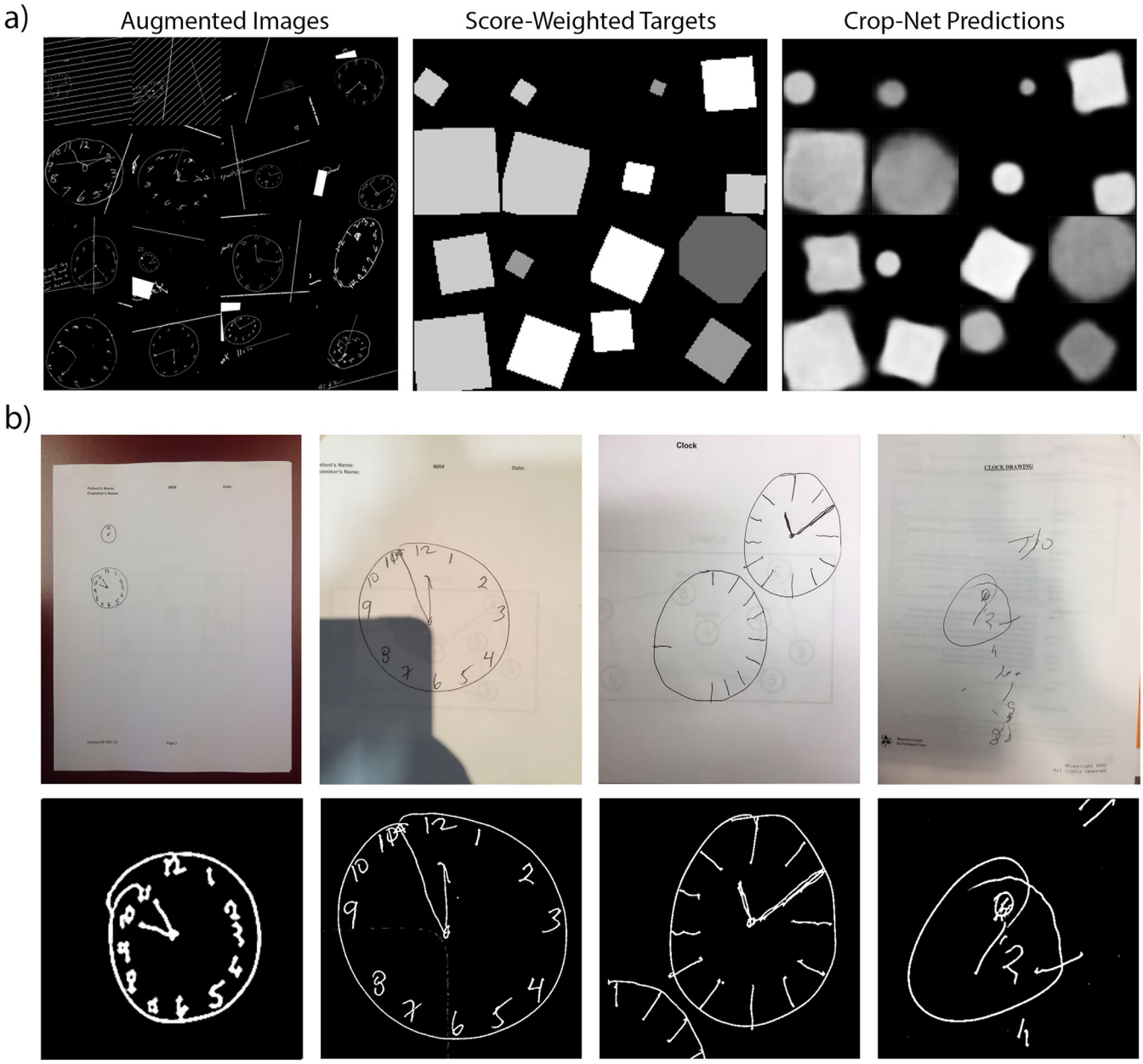

FIGURE 2 Preprocessing examples. A, Augmented training data and Crop-Net's score-weighted output compared to the target. B, Examples of image preprocessing and cropping drawn from the Toronto Dementia Research Alliance dataset. The original images are in the first row and the processed images are in the second row

2.4 | ViT clock features for dementia classification

2.4.1 | ViT model

To generate clock features useful for predicting dementia diagnosis, we used the PyTorch torchvision implementation of the vision transformer (ViT) "vit_l_16" architecture from Dosovitskiy et al.[12] with the "ViT_L_16_Weights.IMAGENET1K_SWAG_E2E_V1" pretrained weights.[19] The head of the model was removed and replaced with a small fully connected network designed to output the clock score ("cg < round > dclkdraw"; scored from 0–5) and previous dementia/Alzheimer's disease diagnosis as reported by the respondent ("hc < round > disescn9"; with "yes" and "previously reported" set to 1, 0 otherwise) from the NHATS dataset. The network comprised three fully connected layers (1024 input and output features) with the leaky ReLu activation function (0.2 negative slope) followed by one fully connected layer (1024 feature input and 2 feature output) with the sigmoid activation function. The clock output (0–1) was multiplied by 5 to match the score range, that is, 0 to 5 (the dementia output was not modified). The model is depicted in Figure 3.

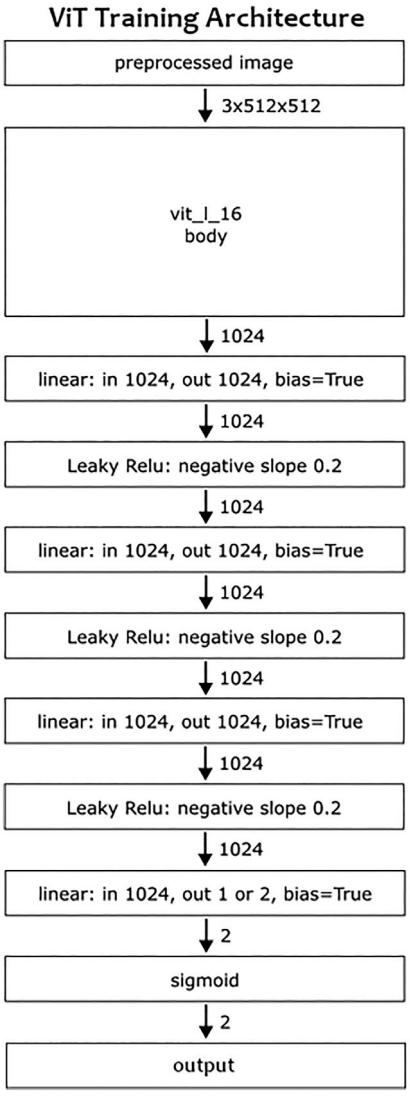

FIGURE 3 ViT classification architecture for training. The single-channel preprocessed image was converted to 3-channel to conform with the expected input of vit_l_16. The head of vit_l_16 was replaced with the fully connected linear layers depicted in the diagram. The output of the network was two-dimensional, representing clock score and reported dementia diagnosis. Outputs are between 0 and 1. ViT, vision transformer

2.4.2 | ViT model training

NHATS data were divided into training (47,485 samples) and validation (1000 samples) sets. Both sets were balanced by replicating underrepresented scores in the training set for equal samples per score (0–5), accounting for the prevalence of high scores in the non-clinical elderly NHATS population. Clock images were preprocessed and cropped as outlined in section 2.3. To improve generalizability, data augmentation was applied to the preprocessed clock images during training and validation, with scaling (0.7–1 times original size), translation (by 0–0.5 times the respective dimension size in all directions), rotation (−180 to 180 degrees), and shearing (−10 to 10 degrees; all uniformly sampled).

With MSE as the loss function, the Adam algorithm was used with learning rate = 0.00002, betas = (0.5,0.999), and all other parameters set to default. The learning rate was divided by 5 when the validation MSE plateaued on the validation data. To terminate the learning process, early stopping was applied when the MSE subjectively remained plateaued after decreasing the learning rate.

The head was trained first, with the weights of the body (vit_l_16) frozen, until MSE plateaued on the validation data (50 batch size). The body weights were then unfrozen, and training continued until MSE plateaued on the validation data (4 batch size with gradients accumulated over 10 iterations due to VRAM constraints). The validation MSE over training is depicted in Figure S1 in supporting information.

2.4.3 | ViT features

After training was complete, the head was removed, and the 1024-dimensional output of the finetuned vit_l_16 body was used as the ViT features.

2.5 | Clock features for comparison methods

All methods, except the human-scored approach, were trained and tested on preprocessed and cropped clock images as outlined in section 2.3.

2.5.1 | Human-scored features

The human-scored clock drawing features comprise 15 binary variables scored by trained individuals in a TDRA memory clinic. Each feature was scored as 1 or 0, depending upon whether the criterion for the feature was satisfied or not (see Table S2 in supporting information for the set of features).

2.5.2 | MiniVGG features

The MiniVGG features are based upon the CNN method used in Sato et al.[10] The network architecture outlined in Sato et al.[10] was used, with the exception that the softmax in the output layer was replaced with the sigmoid function to match the training approach used for ViT. As was done with ViT, the network was trained to predict the clock score and reported dementia diagnosis on the NHATS data, except that the body and head were trained simultaneously from the start because pretrained weights were not used. After training was complete, the final layer was removed, and the 512-dimension output was used as the MiniVGG features.

2.5.3 | MNv2 features

The MNv2 features are based upon the CNN method outlined in Amini et al.,[9] using a PyTorch torchvision implementation of the MNV2

TABLE 1 Dementia and normal cognition group demographics.

Groups	Both groups	Dementia	Normal cognition	Dementia—normal cognition
Number of participants	862	522	340	182
Female %	55.6	50.6	63.2	−12.6***
Average age	70.1 (13.2)	75.5 (10.9)	62.0 (12.2)	13.5***
Average years of education	14.7 (3.8)	13.9 (3.9)	16.0 (3.3)	−2.1***
MMSE total score	26.7 (3.5)	24.3 (3.3)	29.2 (1.2)	−4.9***

Note: Includes number of participants, percentage of female participants, age, years of education (YOE), and MMSE scores for both groups together and separately, as well as group differences. Standard deviations in parentheses.
*** = P(groups are equivalent) < 0.001; calculated using bootstrap analysis.
Abbreviation: MMSE, Mini-Mental State Examination.

architecture[15] with the "MobileNet_V2_Weights.IMAGENET1K_V2" pretrained weights (https://pytorch.org/vision/main/models/generated/torchvision.models.mobilenet_v2.html). No additional training was performed. The fully connected layer was removed and replaced with a global average pooling layer, reducing the 7 × 7 spatial dimensions to 1. The resultant 1280-dimension output was used as the MNv2 features.

2.5.4 | RF-VAE features

The RF-VAE features are based upon the RF-VAE method used in Bandyopadhyay et al.[11] The 10 features output by the pretrained encoder (https://github.com/iheallab/Clock-Drawing-Classification-With-RF_VAE) were used as the RF-VAE features.

2.6 | Dementia classification on the TDRA dataset

2.6.1 | Dementia and normal cognition groups

For this study, dementia diagnosis (n = 522) included those diagnosed by TDRA clinicians with Alzheimer's disease, dementia with Lewy bodies, dementia NYD (cause not yet determined), and vascular dementia.

The normal cognition group (n = 340) included those diagnosed as "subjective cognitive decline/subjective cognitive complaints (SCC; patient has complaints but normal cognitive testing)" or "normal cognition—baseline" in the TDRA dataset. Basic demographic information and Mini-Mental State Examination (MMSE) scores[20] for both groups are listed in Table 1.

2.6.2 | Dementia versus normal cognition classification from clock drawing features

Linear regression was used to predict dementia (coded as 1) versus normal cognition (coded as 0) from clock drawing features produced using our current method, ViT, and four comparison methods: Human, MiniVGG, MNv2, and RF-VAE.

To improve generalizability to clock image variations that could occur as a result of the imaging process (e.g., photographing the clock) or image preprocessing, the TDRA training dataset was augmented with 50 permutations of each clock drawing by randomly scaling (0.7–1 times original size), translating (by 0–0.5 times the respective dimension size in all directions), rotating (−45 to 45 degrees), and shearing (−10 to 10 degrees; all uniformly sampled) the images for three deep learning methods (ViT, MiniVGG, MNv2). The augmentation was not performed for RF-VAE because the model was trained on digital clock drawings that were not subject to such sources of variability (consequently, the RF-VAE linear model performed worse when trained using the 50 permutations: receiver operating characteristic [ROC] area under the curve [AUC] 0.730 vs. 0.745).

For the Human and RF-VAE features (15 and 10, respectively), linear regression models were trained using leave-one-out cross-validation to estimate out-of-sample performance. For the ViT, MiniVGG, and MNv2 features, due to the high dimensionality of the features relative to the dataset sample size (1024, 512, and 1280 vs. 862), the feature dimensionality was reduced to 100 using principal component analysis (PCA) trained on the NHATS dataset. Lasso linear regression models were then trained using nested cross-validation, with leave-one-out cross-validation in the outer loop and 5-fold cross-validation in the inner loop to select the penalty value (α), from $1/10^3$ to 1/10 in exponent increments of 0.2.

2.6.3 | Classification performance

We evaluated the performance of all five methods using multiple metrics: area under the ROC curve (AUC), balanced accuracy (maximized over the decision threshold), F1 score (maximized over the decision threshold), and true positive rates (TPRs) at fixed false positive rates (FPRs) of 5% and 10%, respectively.

ViT is statistically compared to the other four methods using paired-samples bootstrapping (1000 samples) to generate P(comparison model ≥ ViT).

TABLE 2 Dementia prediction measures of performance.

Method	ROC AUC	Balanced accuracy %	F1 score	% TPR at 5% FPR	% TPR at 10% FPR
ViT	**0.835**	**76.5**	**0.813**	**49.8**	**61.5**
Human	0.771 (0.000 ***)	74.3 (0.031 *)	0.782 (0.006 **)	46.2 (0.248)	54.2 (0.048 *)
MiniVGG	0.783 (0.000 ***)	73.3 (0.008 **)	0.787 (0.017 *)	34.1 (0.003 **)	46.9 (0.000 ***)
MNv2	0.777 (0.000 ***)	72.3 (0.000 ***)	0.795 (0.041 *)	29.5 (0.000 ***)	40.2 (0.000 ***)
RF-VAE	0.745 (0.000 ***)	69.1 (0.000 ***)	0.769 (0.000 ***)	27.8 (0.000 ***)	40.6 (0.000 ***)

Note: Measures shown for our current model (ViT; in bold), a model based on human-scored features (Human), a model based on Sato et al.[10] (MiniVGG), a model based on Amini et al.[9] (MNv2), and a model based on Bandyopadhyay et al.[11] (RF-VAE). The thresholds for balanced accuracy and the F1 scores are those that maximized the respective measure for the given method. Scores with the best performance between models are in bold. Models sorted by balanced accuracy. Numbers in brackets are P(model > = ViT) for the respective measure (*** = $P < 0.001$; ** = $P < 0.01$; * = $P < 0.05$; · = $P < 0.1$).
Abbreviations: FPR, false positive rate; MNv2, MobileNetV2; RF-VAE, relevance factor variational autoencoder; ROC AUC, receiver operating characteristic area under the curve; TPR, true positive rate; ViT, vision transformer.

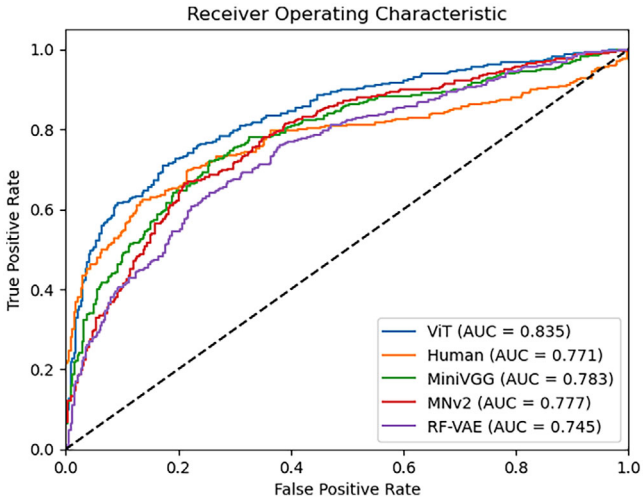

FIGURE 4 Dementia prediction ROC curves. Curves depicted for our current model (ViT), a model based on human-scored features (Human), and three comparison models (MiniVGG, MNv2, RF-VAE). AUC, area under the curve; MNv2, MobileNetV2; ROC, receiver operating characteristic; RF-VAE, relevance factor variational autoencoder; ViT, vision transformer

3 | RESULTS

3.1 | TDRA dementia prediction

Using linear models, we trained, validated, and tested the clock drawing features produced using our current approach, ViT, and four comparison methods (Human, MiniVGG, MNv2, RF-VAE) in the context of predicting dementia ($n = 522$) versus normal cognition ($n = 340$) in the TDRA clinical dataset ($n = 862$).

The results for each method are summarized in Table 2, and the ROC curves are displayed in Figure 4. In Table 2, we report ROC AUC, balanced accuracy (maximized over the decision threshold), F1 score (maximized over the decision threshold), and TPRs at fixed FPRs of 5% and 10%, respectively. For each measure, we statistically compare ViT's performance to the four comparison methods, reporting the probability of the null hypothesis, P(comparison method ≥ ViT), in brackets.

ViT outperformed human and other automated methods in predicting dementia. Moreover, ViT's superior performance was statistically significant in all cases except compared to human-scored features for TPR at 5% FPR ($P = 0.248$).

4 | DISCUSSION

We developed a deep neural network (DNN) model for detecting dementia from CDT photographs, with 76.5% balanced accuracy, outperforming human-scored features and prior methods on the TDRA dataset. Our approach: (1) automates preprocessing and cropping of photographs with poor lighting and markings, (2) demonstrates generalizability by training on a large diverse dataset and testing on a separate clinical dataset, and (3) was directly compared on the same dataset to human-scored features and other automated methods using consistent classification methodology.

ViT outperformed human-scored features on all accuracy measures for predicting dementia from CDT photos (Table 2). While the Freedman system[14] performed well and is reliable,[21,22] ViT's superior accuracy and automation benefits (e.g., consistency, scalability, cost effectiveness) justify its use over manual methods. However, because clinicians typically use qualitative assessment rather than feature scores, future studies should compare ViT to clinicians' diagnoses based on CDT images.

ViT also outperformed prior methods for dementia classification from CDT photos/scans. For comparison, we extracted features from MiniVGG,[10] MNv2,[9] and RF-VAE,[11] and then trained and tested on the same TDRA dataset. Our results show that linear models are sufficient when provided high-quality features, and ViT extracted better features than other methods.

A challenge in applying DNNs to the CDT is preprocessing/cropping images that may contain irrelevant markings or multiple drawing attempts. Our approach combines line filtering with a custom CNN designed to locate the most clock-like drawing regardless of size, orientation, position, or quality, thereby improving the input quality for subsequent classification. However, our method is not flawless. While it occasionally struggles to isolate a single clock when multiple drawings appear in proximity (4/522 dementia samples versus 1/340 normal cognition samples), this limitation may be informative, as multiple drawing attempts could themselves indicate cognitive impairment.

ViT's superior performance compared to the CNN-based and VAE methods likely stems from its architectural advantages, namely its capacity via self-attention mechanisms to capture global dependencies, long-range interactions, and overall clock structure,[23] in contrast to the locally focused operations common in CNNs and the VAE's encoder/decoder. Additionally, ViT's larger input image size (512 × 512) compared to MiniVGG (32 × 32), MNv2 (256 × 256), and RF-VAE (64 × 64) may also contribute to its effectiveness. Future work should investigate the impact of such factors on DNN-based feature extraction to optimize diagnostic accuracy and computational efficiency.

A central element of our approach was that the features extracted by ViT were learned to better predict clock score and dementia diagnosis. An alternative approach, pursued in previous research,[6,7,24] would be to train models to extract a predefined set of relevant clock features, for example, shape of the contour, the angle, location and length of the clock hands, and the identity and location of the clock numbers. This predefined feature extraction method offers greater transparency and easier integration into existing scoring systems. However, it also introduces more potential points of failure, increases the risk of researcher bias, and may fail to capture more subtle or holistic aspects of the clock image relevant to dementia detection, such as the shape or symmetry of clock elements. This concern is supported by our findings, which show that the features extracted by ViT outperformed 15 predefined human-scored features in diagnostic accuracy.

Another alternative to our approach is using a digital pen and paper for clock drawings, with a device like a tablet capturing additional hand movement data (pressure, velocity, acceleration, trajectory) to reflect cognitive and motor abilities, accurately predicting cognitive status.[8] However, this requires specific devices that may be unavailable in remote or low-resource settings, and elderly test takers less familiar with digital tools might perform worse, potentially skewing results and reducing diagnostic accuracy. Additionally, this method may not be compatible with existing applications of the CDT, including a large proportion of past CDT data, for example, the TDRA dataset.

In this study, we used the CDT to predict dementia without incorporating other cognitive tests or demographic factors like age, sex, education, or family history, which could improve accuracy.[25] For instance, the Benson Complex Figure Test could evaluate visuospatial function and visual memory, which are frequently impaired in Alzheimer's disease.[26] Future studies should explore combining multiple automated diagnostic sources, including demographics, to enhance the ViT-based approach's performance. Additionally, adapting this ViT-based framework for automated analysis of other drawing tasks, such as the Rey–Osterrieth complex figure test, represents a valuable future direction, although this would likely require tailored preprocessing and model retraining.

While this study demonstrates the strength of the ViT approach in distinguishing between normal cognition and dementia using CDT images, it is important to acknowledge the scope of the current classification. The binary distinction (normal cognition vs. dementia) does not address the identification of the earlier stages of cognitive decline, such as mild cognitive impairment (MCI), that facilitates timely intervention.[27,28] Our initial focus on a clear diagnostic contrast was instrumental in validating the core methodology and demonstrating the potential of ViT-extracted features. Future research should investigate the capability of this ViT-based framework to differentiate MCI from both normal cognition and dementia, thereby addressing the need for early screening and preventative cognitive health-care strategies.

In conclusion, our study highlights ViT's potential to automatically detect dementia from CDT photographs, outperforming human-scored features and other automated methods on a clinical dataset despite challenging image quality. Our approach could be integrated into an app for virtual dementia screening, providing instant feedback, connecting users to health professionals, and allowing pen-and-paper use—a critical advantage for elderly or cognitively impaired individuals. As aging populations increase the global dementia burden, our method could enhance screening accessibility, reduce costs, and improve lives.

AUTHOR CONTRIBUTIONS

Conceptualization: Michael B. Bone, Morris Freedman, Bradley R. Buchsbaum. Methodology: Michael B. Bone, Morris Freedman, Bradley R. Buchsbaum. Investigation: Michael B. Bone, Morris Freedman, Sandra E. Black, Daniel Felsky, Sanjeev Kumar, Bradley Pugh, Stephen C. Strother, David F. Tang-Wai, Maria Carmela Tartaglia, Bradley R. Buchsbaum. Data Curation: Michael B. Bone, Morris Freedman, Sandra E. Black, Daniel Felsky, Sanjeev Kumar, Bradley Pugh, Stephen C. Strother, David F. Tang-Wai, Maria Carmela Tartaglia, Bradley R. Buchsbaum. Software: Michael B. Bone. Formal Analysis: Michael B. Bone. Writing—original Draft: Michael B. Bone. Writing—review and editing: Michael B. Bone, Daniel Felsky, Morris Freedman, Bradley R. Buchsbaum. Visualization: Michael B. Bone. Supervision: Michael B. Bone, Bradley R. Buchsbaum.

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CONFLICT OF INTEREST STATEMENT

Michael Bone, Morris Freedman, Bradley Pugh, Daniel Felsky, Stephen Strother, and Bradley Buchsbaum are affiliated with the Baycrest Centre for Geriatric Care, which holds a patent on the automated screening

method described in this report. Michael Bone, Morris Freedman, and Bradley Buchsbaum are authors on the patent. Sandra E. Black, Sanjeev Kumar, David F. Tang-Wai, and Maria Carmela Tartaglia have nothing to disclose. Author disclosures are available in the supporting information.

CONSENT STATEMENT

For the TDRA and NHATS datasets, written informed consent was originally obtained for all participants for the materials to be included in the respective databases. Confirming consent for use in the current study was not necessary.

DATA AVAILABILITY STATEMENT

Data will be made available upon request and ethics board approval. Code will be available on Zenodo upon publication.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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